

N7 goes high and DIS1 shows 'F' indicating that the water tank is full.

When water starts overflowing the tank, pin 13 of gate N6 goes low to make output pin 12. The buzzer sounds

to indicate that water is overflowing the tank and you need to switch off the motor pump.

Assemble the circuit on a general-purpose PCB and enclose in a suitable

box. Use a non-corrosive material such as steel strip for the five sensors and hang them in the water tank as shown in the circuit diagram. Use regulated 5V to power the circuit. ●

EMERGENCY PHOTO LAMP

■ RAJU BADDI

This emergency light can be powered either by a rechargeable battery (like 3.6V Ni-Cd) or a non-rechargeable battery (3.0V CR2032). The white LED (LED1) glows automatically when the power fails and you are left in dark. The quiescent current of the circuit is very low and

the battery is practically used only when the LED glows.

Fig. 1 shows the circuit of the emergency photo lamp. When light falls on phototransistor T1 (L14F1), it conducts and transistor T2 does not conduct, making output of gate N1 high. As gates N2, N3 and N4 are connected to the output of gate N1, their outputs also remain high. As a result, the white LED does not glow. An additional circuit for 5V DC socket is shown within the dotted lines for charging

the rechargeable battery.

In the dark, when light does not fall on phototransistor T1, it does not conduct and transistor T2 conducts, making the output of gate N1 low. As a result, the outputs of gates N2, N3 and N4 also remain low and the white LED glows.

Assemble the circuit on a small general-purpose PCB and fit it (including the battery) inside the cap of a gluestick tube with the LED facing down (see Fig. 2). Hang it suitably at your workplace, say, in the office or kitchen, such that the ambient light falls directly on the phototransistor. ●

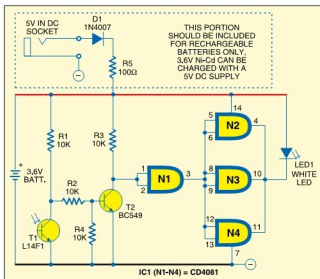


Fig. 1: Emergency photo lamp circuit

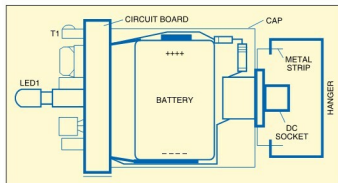


Fig. 2: Proposed assembled unit

STANDBY POWER-LOSS PREVENTER

■ D. MOHAN KUMAR

Electronic devices consume some power even in the standby mode, i.e., when they have been switched off using a remote handset but not the mains power switch. For instance, when a CRT TV or PC moni-

tor is in use, it consumes 80-100 watts of power. In the standby mode too, it draws a few watts of power. Thus if you leave these devices in standby mode for a long time, they may inflate your electricity bill.

The circuit described here helps you save on your electricity bill by

disconnecting the device (say, CRT TV) from the mains power automatically